

Worksheet — 2.3 Algorithms

Name: ___ Date: ____

OCR A Level Computer Science H446, Component 02, Section 2.3.1 — Algorithms.

Time: ~60 min. SHOW YOUR TRACES.

Section A — Skills practice

Activity 1 — Big O match

Match each algorithm to its **best**, **average** and **worst** time complexity by writing the correct Big O in each cell. Choose from: $O(1)$, $O(\log n)$, $O(n)$, $O(n \log n)$, $O(n^2)$.

Algorithm	Best	Average	Worst
Linear search			
Binary search			
Bubble sort (with swap flag)			
Insertion sort			
Merge sort			
Quick sort			

Activity 2 — Trace a sort (bubble OR insertion)

Sort this list into **ascending** order: [5, 1, 4, 2, 8]

Circle which you are tracing: **BUBBLE / INSERTION**

Fill in the list state **after each pass**. Add rows if you need them.

Pass	List state after this pass	Swaps this pass
Start	5 1 4 2 8	—
1	— — — — —	
2	— — — — —	
3	— — — — —	
4	— — — — —	

Final sorted list: _ _ _ _ _

Activity 3 — Trace a binary search

Sorted list (index 0–8):

Index	0	1	2	3	4	5	6	7	8
Value	3	7	11	15	19	23	27	31	35

Search for 27. Use `mid = (low + high) DIV 2` (integer division, round down). Fill the table.

Step	low	high	mid (index)	value at mid	< = > target?
1					
2					
3					
4					

Found at index: Number of comparisons:

Activity 4 — Big O justification (short answer)

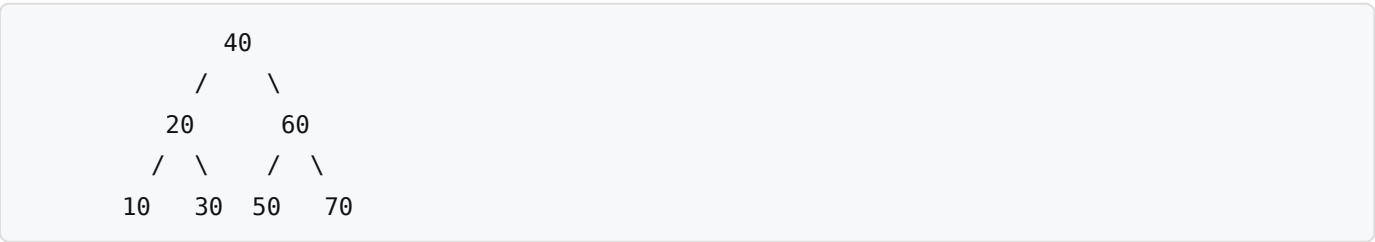
(a) Explain why binary search is **$O(\log n)$** . Refer to what happens to the list each comparison.

(b) Explain why bubble sort is **$O(n^2)$** in the average case.

(c) State the precondition binary search requires that linear search does not.

Activity 5 — Tree traversals

Binary tree (text form):



Write the output of each traversal:

Pre-order (Node → Left → Right):

In-order (Left → Node → Right):

Post-order (Left → Right → Node):

Which traversal outputs a BST in ascending order? __

Activity 6 — Dijkstra's shortest path

Weighted, undirected graph given as an **edge list** (node1 — node2 : weight):

A — B : 4
A — C : 2
B — C : 1
B — D : 5
C — D : 8
C — E : 10
D — E : 2
D — F : 6
E — F : 3

Find the shortest path and total distance from A to F.

Working table — write the tentative distance to each node, updating as you settle nodes. Cross out / overwrite when a shorter route is found.

Settled node (order)	A	B	C	D	E	F
Initial	0	∞	∞	∞	∞	∞

Shortest path A → F: → → → →

Total distance: _____

Challenge

Challenge box. A* search uses $f(n) = g(n) + h(n)$.

(a) For the graph above, suppose you are given a straight-line heuristic estimate to F of: $h(A)=9$, $h(B)=7$, $h(C)=8$, $h(D)=4$, $h(E)=3$, $h(F)=0$. Is this heuristic **admissible**? Explain how you would check, and what admissible means.

(b) State ONE situation where A explores fewer nodes than Dijkstra, and ONE condition under which A behaves exactly like Dijkstra.

...

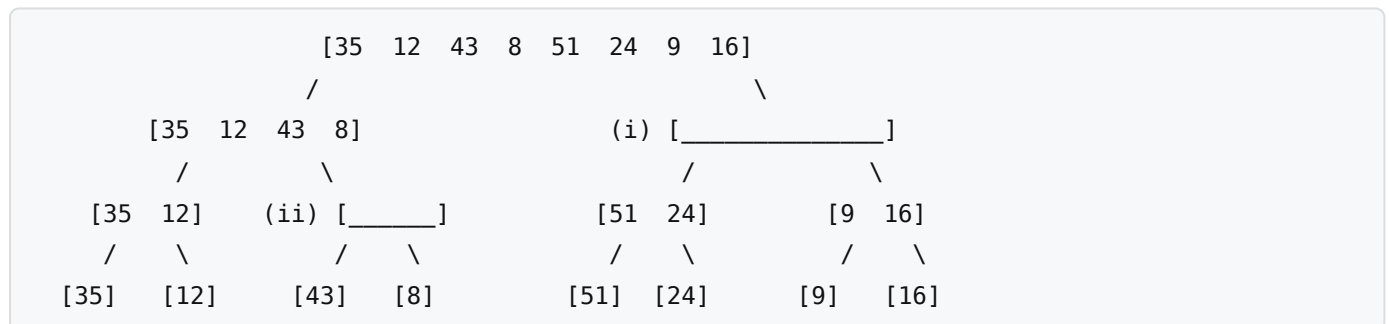
...

Activity 7 — Merge sort: complete the divide and merge diagram

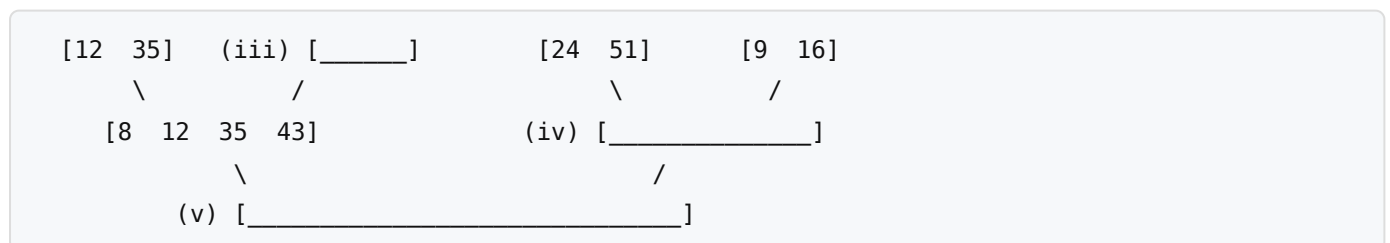
Merge sort the list [35, 12, 43, 8, 51, 24, 9, 16].

(a) Complete the missing boxes (i)–(v) in the diagram.

DIVIDE (halve repeatedly until every sublist has one element):



MERGE (combine pairs back together, in order, comparing front elements):



(b) For this list of 8 items, how many **levels of dividing** are needed to reach single-element lists? ____ **How is this number related to $n = 8$?** ____

(c) Approximately how much work (comparisons/copies) is done at **each merge level**, in terms of n ? ____ Hence state why merge sort is **$O(n \log n)$** .

(d) Unlike quicksort, merge sort's worst case is still $O(n \log n)$. Give the **cost** merge sort pays for this reliability. _____

Activity 8 — A* vs Dijkstra: which and why?

Weighted, undirected graph (edge list) and a heuristic estimate $h(n)$ of the distance from each node to the goal **G**:

S – A : 1	S – B : 4	S – D : 2
A – B : 2	A – C : 5	
B – C : 2	C – G : 3	

Node	S	A	B	C	D	G
$h(n)$	7	6	4	3	9	0

(a) **Dijkstra from S**. Complete the table (settle the unvisited node with the smallest distance each round) and write the **order the nodes are settled**.

Settled (order)	S	A	B	C	D	G
Initial	0	∞	∞	∞	∞	∞

Settle order: _ $\rightarrow$ _ $\rightarrow$ _ $\rightarrow$ _ $\rightarrow$ _

(b) **A* from S to G**. Each time you expand a node, record it with its g , h and $f = g + h$. Always expand the open node with the **lowest f**. Stop when G is expanded.

Node expanded	$g(n)$	$h(n)$	$f(n)$
S	0	7	7

Shortest path $S \rightarrow G$: _____ $\rightarrow$ _____ $\rightarrow$ _____ $\rightarrow$ _____ $\rightarrow$ _____ Distance: _____

(c) Which node does **Dijkstra settle** but **A* never expands**? _____ Use its f value to explain why A* ignores it.

(d) Decision. A satnav needs the route from one start to **one destination** on a map where straight-line distances are known. A network-monitoring tool needs the shortest routes from one server to **every** other node. State, with a reason, which algorithm suits each task.

Section B — Exam practice (30 marks)

Answer all questions. Questions marked with an asterisk (*) assess the quality of your extended response. All code must be written in OCR Exam Reference Language.

Q1. State the **worst-case** time complexity of quicksort in Big O notation, and describe a situation in which this worst case occurs. **[2]** (AO1)

Q2. The following procedure is written in OCR Exam Reference Language, where **n** is the number of elements in the array **arr**.

```
procedure pairs(arr:array, n)
  for i = 0 to n - 1
    for j = 0 to n - 1
      print(arr[i] + arr[j])
    next j
  next i
endprocedure
```

Determine, with justification, the **time complexity** of this procedure in Big O notation. **[3]** (AO2)

Q3. The following algorithm searches the sorted array **arr = [4, 9, 15, 22, 30, 41, 56, 70]** (indexes 0–7) for **target = 10**.

```

low = 0
high = 7
found = false
while low <= high AND found == false
    mid = (low + high) DIV 2
    if arr[mid] == target then
        found = true
    elseif arr[mid] < target then
        low = mid + 1
    else
        high = mid - 1
    endif
endwhile
if found then
    print(mid)
else
    print(-1)
endif

```

(i) Complete the trace table, adding one row for each iteration of the while loop. **[3]** (AO2)

low	high	mid	arr[mid]	found
0	7			

(ii) State the value printed when the algorithm ends. **[1]** (AO2)

Q4. A sorted array `arr` contains `n` integers (zero-indexed). Write a function `searchIndex(arr, n, target)` using **OCR Exam Reference Language** that performs a **binary search** and returns the index of `target` in `arr`, or `-1` if it is not present. **[5]** (AO3)

.....
.....
.....
Q5. State the data structure used to implement:

(i) a **breadth-first** traversal of a graph. [1] (AO1)

.....
.....

(ii) a **depth-first** traversal of a graph. [1] (AO1)

.....
.....

Q6. A graph is stored as the following adjacency list:

```
A : B, C
B : A, D
C : A, E
D : B, E
E : C, D
```

(i) A breadth-first traversal starts at A and visits neighbours in alphabetical order. State the order in which the nodes are visited. [2] (AO2)

.....
.....
.....

(ii) The edges of this graph are unweighted. Explain why a breadth-first traversal from A finds a path to E using the **fewest edges**, while a depth-first traversal may not. [1] (AO2)

.....
.....

Q7. The A *pathfinding algorithm* assigns each node a value $f(n) = g(n) + h(n)$. State what $g(n)$ and $h(n)$ each represent. [2]* (AO1)

.....
.....
.....

Q8* A music streaming service stores 50 million track records, held in order of track ID. Users search for tracks by ID thousands of times per second. Each night a batch of a few thousand new tracks is added and the full file must be put back in order; the nightly process runs on a server with a large amount of memory. A cut-down version of the software must also run on an embedded car system with very limited memory, where it keeps a few hundred downloaded tracks in order.

Discuss which **searching and sorting algorithms** the developers should choose for this system, justifying your choices with reference to **time and space complexity**. [9] (AO3)

[illegible]